Timetable, official real time or Bide+? A paired comparison of bus arrival predictions on the same buses, stops and minutes in the Bizkaibus network

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Abstract

Passengers of many bus networks receive real-time waiting times from the operator, either directly or through apps that relay them, but the accuracy of these predictions is rarely compared with alternatives on the same buses. Bizkaibus, the interurban bus network of Biscay (Spain), shows its own waiting times in an official app and publishes, as open data, the positions of its vehicles and its timetable. For one week in November 2026 we recorded the official waiting times of 60 stops every minute and the live positions every 20 seconds, and compared three predictions for the same bus at the same stop and minute: the published timetable, the official real-time announcement and the arrival predicted by the Bide+ method, which uses only the open data and is documented in a published technical report, all against the actual passage reconstructed from the feed. Over 160,046 paired predictions on five normal days, the timetable is within two minutes of the actual passage in 16% of cases, the official real time in 36% and Bide+ in 64%. The two real-time sources are both accurate when the bus is less than three minutes away (86% and 88% within two minutes). Further ahead, Bide+ is far more accurate: with the bus 6 to 10 minutes away, 71% of its predictions are within two minutes against 37% of the official ones, and 61% against 27% at 10 to 20 minutes. The official errors are mostly late announcements: in about half of the announcements made 10 to 30 minutes ahead, the bus arrives more than two minutes earlier than announced. Bide+ is more accurate at 58 of the 60 stops, at every hour and on every day. The capture program, the data and the analysis are openly available.

Keywords: bus arrival prediction, real-time passenger information, operator

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predictions, paired evaluation, SIRI, GTFS, open data, reproducibility

1. Introduction

Real-time arrival information shortens the time passengers wait at stops and makes that wait less uncertain (K. E. Watkins et al. 2011; Brakewood and K. Watkins 2019), but only when it is accurate: wrong predictions erode trust and make passengers miss the bus they were told to expect (Gooze et al. 2013; Rahman et al. 2013). Most passengers receive these predictions from the operator itself, through its app and stop displays, or from journey planners that relay the operator's figures. Whether a third party that has only the operator's *open* data can predict better than the operator is rarely measured, and almost never on the same buses and at the same moments.

This paper makes that measurement for Bizkaibus, the interurban bus network of the province of Biscay (Spain). Since 2023 Bizkaibus shows its own real-time waiting times in an official app and on some stop displays. It also publishes, as open data, the GPS position of every vehicle in service (a SIRI Vehicle Monitoring feed, CEN 2022) and its timetable (GTFS, MobilityData 2026), but not its predictions. The *Bide+* method[2] predicts arrivals from those two open sources alone. It is documented in a technical report (Garrido 2026a) and specified for reimplementation in a companion paper (Garrido 2026b). Both evaluated it against the published timetable, which is a weak baseline on this network: after the first hour of a trip, most buses pass a stop more than two minutes before the published time.

Here the baseline is the strongest one available: the operator's own live prediction. The research question is:

> *For the same bus, at the same stop and at the same minute, how close to the actual passage are the published timetable, the real-time waiting time announced by Bizkaibus and the arrival predicted by the Bide+ method from open data?*

For one week in November 2026 we recorded, every minute, the waiting times that the official service gave for 60 stops, and the live feed every 20 seconds. We then paired each official announcement with the bus it referred to, with that bus's published time at the stop, with the Bide+ prediction

[2]Used by the independent Bide+ app, `https://www.bideplus.eus` (formerly Bizkaibus+). The app is not affiliated with Bizkaibus, the Diputación Foral de Bizkaia or the operators of the service.

for that bus at that minute, and with the moment the bus actually passed. Figure 2 summarises the answer. The timetable is within two minutes of the actual passage in about one case in six. The official real time is close to the bus in the last three minutes but falls to about one in three further ahead. Bide+ is within two minutes about twice as often as the official real time, and four times as often as the timetable.

The contributions are:

1. A paired protocol to compare a third-party predictor with an operator's real-time predictions when the operator publishes positions but not predictions: what to capture, how to match each announcement to a vehicle, how to reconstruct the actual passage, and how to avoid giving either side an advantage.
2. The comparison of the three sources, over 160,046 paired predictions on five normal days, by time to arrival, direction of the error, stop, time of day and day, with confidence intervals that respect the dependence between predictions of the same trip, plus a strike day analysed separately.
3. The capture program, the raw data and the analysis, openly available, so that the comparison can be repeated on other days or networks.

Section 2 describes the three sources, Section 3 related work, Section 4 the data, Section 5 the comparison, Section 6 the results and Section 7 their meaning and limits.

2. The three sources

2.1. The published timetable

Bizkaibus publishes its timetable as GTFS. As shown in the companion paper, its intermediate times are interpolated at constant speed between the terminals, not observed: buses are slower in towns and faster on open roads, so after the first hour of a trip most of them pass a stop more than two minutes *before* the published time (Garrido 2026b). The timetable is what a passenger sees in the printed and online timetables and in any app without real-time data. Here its prediction for a bus is simply the published time of its trip at the stop.

2.2. The official Bizkaibus real-time waiting times

The official Bizkaibus app (since 2023) and some stop displays show, for each stop, the waiting time of the next buses of each line. The app

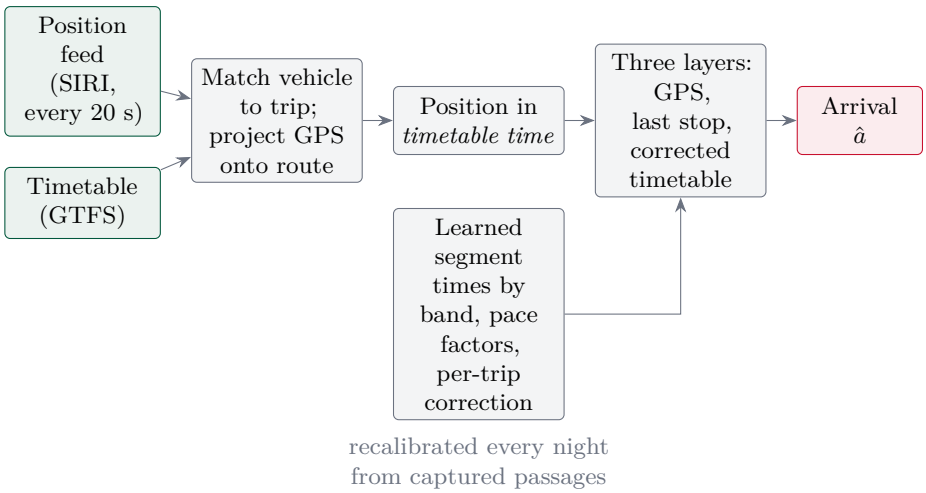


Figure 1: The Bide+ method (Garrido 2026a; Garrido 2026b). Only open data enter it: the operator's position feed and timetable.

obtains them from a public web service (`GetPasoParadaMobile_JSON`) that, for a stop, returns the next two buses of every line that serves it, each with its waiting time in whole minutes and its distance in metres. How these times are computed is not published. They are the figures that a Bizkaibus passenger sees when checking the official sources.

2.3. The Bide+ method

The Bide+ method (Garrido 2026a; Garrido 2026b) predicts the arrival of each bus at each stop from the public position feed and the timetable. Its design responds to two properties of the Bizkaibus data: the feed reports positions but no delays, and the intermediate times of the timetable are interpolated at constant speed between the terminals, so they drift further from the actual passage as the trip advances. Figure 1 outlines it.

In brief:

- **Timetable time.** Each vehicle is matched to its scheduled trip and its GPS position is projected onto the route and converted into the scheduled time at that point. Predictions are then built from *differences* of timetable times, which are reliable even when the absolute intermediate times are not.
- **Learned segment times.** The actual travel time between each pair of consecutive stops is learned from captured passages by time band,

as a median shrunk towards the all-day median of the segment, with a fallback by segment length for segments without data.

- **Three layers.** The prediction uses, in order, the projected GPS position, the last stop reported by the feed and the timetable corrected trip by trip. Each layer abstains under explicit rules, and the state of the prediction says which layer produced it.

- **Daily recalibration.** Every night the tables are recalibrated from the latest captures, and a change is adopted only if it reduces the error on held-out data.

Against the published timetable, over 9.1 million predictions in 23 days, the arrival shown by the method was within ±2 minutes in 92–96% of cases with the bus less than three minutes away and in 67% at 10 to 20 minutes, against 23–25% for the timetable (Garrido 2026b). The report (Garrido 2026a) gives the measurements behind each constant and the history of each rule. An open reference implementation in Python reproduces the production engine exactly (it passes 407 test vectors produced by that engine), and it is the implementation used in this paper.

3. Related work

The accuracy of operator real-time information has been studied mostly with the operator's own records. Cats and Loutos (2016) evaluated the bus arrival information of Stockholm against automatic vehicle location data and found that errors grow with the prediction horizon and depend on the stop and the time of day. Studies of passenger benefits (K. E. Watkins et al. 2011; Mishalani et al. 2006; Liu and Miller 2020) take the operator's predictions as given and show that their value depends on their accuracy (Gooze et al. 2013). On the other side, predictors built only on public data are feasible: García-Mauriño et al. (2024) improve arrival predictions using only data available through public transport APIs, and Wessel et al. (2017) reconstruct actual passage times from a public vehicle location feed, the kind of ground truth used here. Reviews of bus arrival prediction (Moreira-Matias et al. 2015; Singh and K. Kumar 2022; B. A. Kumar et al. 2025) compare methods with each other or with the timetable, rarely with the operator's live service. This paper fills that gap: operator predictions and an open-data predictor on the same vehicles, stops and moments, against the same actual passage, with the operator's predictions recorded as passengers saw them.

The evaluation follows the rules of out-of-sample forecast evaluation (Tashman 2000): the Bide+ method is calibrated only with data from before the week that is measured, so nothing from the test days leaks into it (Kaufman et al. 2012).

4. Data

4.1. What was captured

A program running on GitHub Actions captured, from 06:00 to 23:00 Madrid time (at night the feed repeats stale data):

- **The live feed**, every 20 seconds: the public SIRI Vehicle Monitoring file of Bizkaibus, with the GPS position, the trip and the next stop of every vehicle in service. The file changes every 60 seconds, so two of every three copies repeat the previous one.
- **The official waiting times**, about once a minute for each of 60 stops: the full response of the official web service for that stop. The stops are queried one after another, about one query per second in total.
- **The timetable** of the day (GTFS), once a day.

The 60 stops are intermediate stops with many buses and lines, at least 15 minutes after the start of the trip (where predictions are hardest and most useful), plus 28 *priority* stops: stops pointed out by passengers and the Bilbao stops where the actual passage deviates most from the published time. Each stop and the reason for its choice are listed in the repository.

4.2. Days

The week runs from Sunday 8 to Saturday 14 November 2026 (Table 1). Wednesday 11 was a general strike, with minimum services (about 30%): about 160 vehicles in the feed against 240 on a normal day. It is analysed separately and does not enter the results of the week. Monday 9 lacks its morning, Thursday 12 lacks official times after 17:12 (the service stopped answering over plain HTTP; it answers over HTTPS, which the capture has used since) and Saturday 14 starts at 09:41. The live feed has no gap longer than two minutes on any day.

5. Comparison design

5.1. The three predictions

Timetable. The published arrival time $\hat{a}_{\mathrm{tt}}$ of the matched bus's trip at the stop, from the GTFS of the week.

Table 1: Days of the capture (Madrid time). The feed and the official service were captured from about 06:00 to 23:00, except as noted.

Date	Day	Coverage
8 Nov 2026	Sunday	Complete
9 Nov 2026	Monday	12:42–23:00 (morning block lost: the capture overran its time limit)
10 Nov 2026	Tuesday	Complete
11 Nov 2026	Wednesday	Complete, *analysed separately*: general strike, minimum services (about 30%)
12 Nov 2026	Thursday	Feed complete; official times until 17:12 (the service stopped answering over HTTP)
13 Nov 2026	Friday	Complete
14 Nov 2026	Saturday	From 09:41

Official. For a query at time t to stop s, an announcement of m minutes for a bus of line ℓ gives the predicted arrival $\hat{a}_{\text{off}} = t + 60\,m$.

Bide+. The Bide+ prediction is recomputed from the captured feed with the reference implementation, not taken from the live app, so that anyone can repeat it. Its learned tables are those of Saturday 7 November, calibrated with data from before the week, and they are not updated during the week. For the query at t, the Bide+ prediction is the one it gave in the last copy of the feed before t, provided that copy is at most 60 seconds old. It is therefore never more recent than the official one. The main comparison rounds it to whole minutes from t, as the app shows it: $\hat{a}_{\text{B+}} = t + 60\,\max\bigl(0, \text{round}((\hat{a} - t)/60)\bigr)$, so that neither side gains from the resolution of its figures.

5.2. The actual passage

The actual passage a of a bus at a stop is reconstructed from the feed as in the companion paper: it is the midpoint between the last copy of the feed in which the bus had not yet passed the stop and the first copy in which it had, without the bus changing trip in between. Because the feed changes every 60 seconds, a is known to within about ± 30 s. The same a serves both predictions, so this noise is shared and favours neither.

5.3. Matching announcements to buses

The official service does not say which vehicle it announces. For a query at t to stop s, its first announcement for line ℓ is matched to the first bus of line ℓ that actually passes s after t, and its second announcement to the second bus. This is the bus a passenger waiting at s would take. Lines

Table 2: Errors of the three sources by time to the actual passage, five normal days. *Med.*: median absolute error (s). *±2*: share of predictions within two minutes (%). *Early*: share of buses that arrive more than two minutes *before* the predicted time (%). *Timetable*: the published time of the trip at the stop. Bide+ is rounded to whole minutes, as shown to passengers.

		Timetable			Official real time			Bide+		
Minutes	Pairs	Med.	±2	Early	Med.	±2	Early	Med.	±2	Early
0–3	19,329	419	18	54	43	86	14	38	88	9
3–6	18,973	428	17	55	114	52	27	56	77	11
6–10	24,119	442	17	57	167	37	39	68	71	13
10–20	53,865	480	15	60	246	27	51	91	61	16
20–30	43,760	516	14	64	354	19	52	128	48	16
All	160,046	472	16	59	191	36	42	81	64	14

listed twice in the same response are left out, as are announcements whose matched bus passes more than 30 minutes after t or is not tracked by the feed. Figure 4 shows one matched bus.

5.4. Metrics

The error of each prediction is $\hat{a} - a$: positive when the bus arrives *earlier* than announced, the error that makes a passenger miss it. For each of the three sources we report the median absolute error, the share within ±1 and ±2 minutes, the share of buses that arrive more than two minutes early (*early*) and the median signed error, by time to the actual passage $a - t$. Predictions of the same trip are not independent, so confidence intervals for the difference between the two sources are obtained by bootstrap over whole trips (Efron and Tibshirani 1993): each of 400 resamples draws trips (day, line, trip number and vehicle) with replacement, with all their pairs.

6. Results

Over the five normal days analysed (8, 9, 10, 12 and 13 November), 160,046 official announcements were paired with a bus and with the Bide+ prediction for that bus at the same moment: 8,210 passages of 2,648 trips, made by 313 vehicles on 72 lines, at the 60 stops.

6.1. Accuracy by time to arrival

The three sources rank the same way at every horizon (Table 2, Figure 2). The published timetable is within two minutes of the actual passage in only 14–18% of cases, whatever the horizon, with a median error of seven

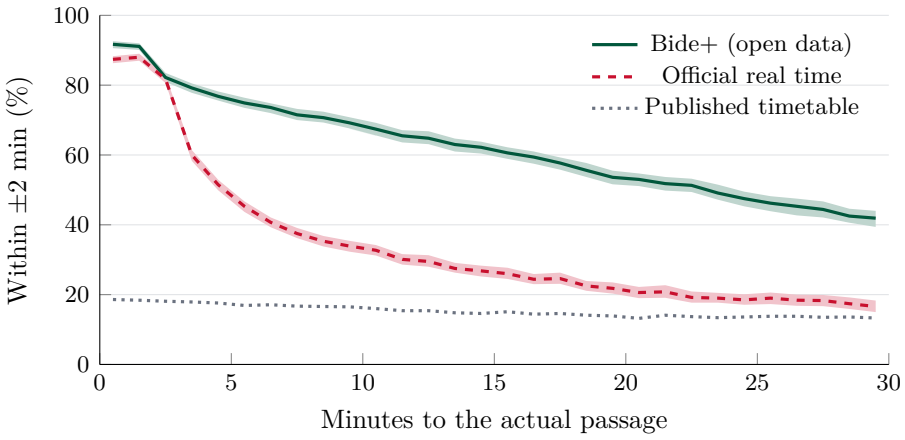


Figure 2: Share of predictions within two minutes of the actual passage, by minutes to that passage: published timetable, official Bizkaibus real time and Bide+ (one-minute bins; shaded: 95% bootstrap interval by trip). Five normal days, 160,046 paired predictions.

to nine minutes. When the bus is less than three minutes away, the two real-time sources are both accurate: 86% of official announcements and 88% of Bide+ predictions are within two minutes. From three minutes on they separate quickly. The official share within two minutes falls to 37% at 6–10 minutes and 27% at 10–20 minutes; Bide+ keeps 71% and 61%. The median error at 10–20 minutes is 246 s for the official service and 91 s for Bide+. The difference in the share within two minutes (Bide+ minus official) has 95% intervals of $[+1.7, +3.7]$ points at 0–3 minutes, $[+23.3, +26.0]$ at 3–6, $[+32.7, +35.8]$ at 6–10, $[+32.6, +36.1]$ at 10–20 and $[+26.5, +30.6]$ at 20–30 minutes. Comparing Bide+ in seconds instead of whole minutes changes these figures by at most one point.

6.2. Direction of the errors

The timetable errs in one direction: buses run ahead of it (median signed error +310 s at 10–20 minutes), and 60% of them pass more than two minutes before the published time. The official real time keeps the same direction with about half the size, and Bide+ is centred (Figure 3). The official median signed error is +124 s at 10–20 minutes and +149 s at 20–30 minutes. In about half of the announcements made 10 to 30 minutes ahead, the bus arrives more than two minutes *earlier* than announced (51–52%, against 16% for Bide+). This is the costly error for a passenger, who plans to reach the stop shortly before the announced time and finds that the bus has gone. Bide+ leans slightly the other way at long horizons (median −24 s at 10–20

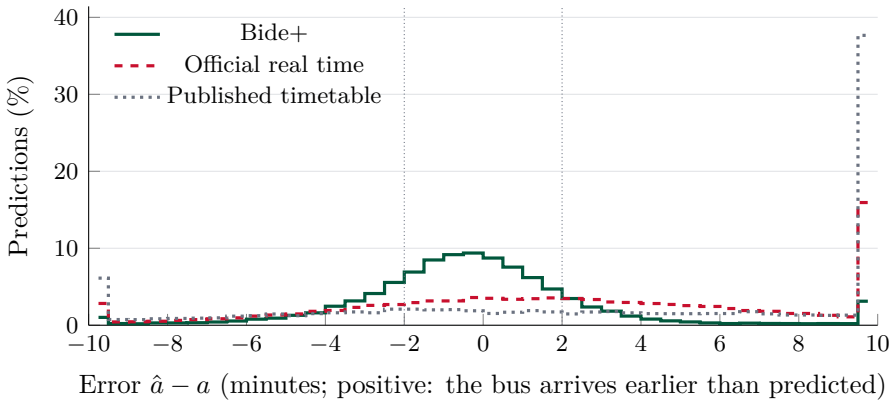


Figure 3: Distribution of the error with the bus 10 to 20 minutes away (30-second bins; the end bins collect everything beyond ±10 minutes). Vertical lines: ±2 minutes. The timetable is shifted far to the right (buses run ahead of it), the official real time less so, and Bide+ is centred and narrow.

minutes and −60 s at 20–30 minutes): when it errs, the passenger more often waits a little longer than misses the bus.

The official announcements are not simply the published timetable: only 12–17% of them are within one minute of the timetable time. They correct a large part of the timetable's error, but not all of it.

6.3. One bus

Figure 4 shows the minute-by-minute announcements for one bus. With 20.8 minutes to go, the official service announced 24 minutes and Bide+ 22; with 9.5 minutes to go, 14 and 10. The official announcement only came within a minute of the actual waiting time in the last five minutes, the pattern behind Figures 2 and 3.

6.4. Stops, time of day and days

The ranking is not confined to a few stops, hours or days. On every day and at every hour the timetable is the least accurate source and Bide+ the most accurate (Figure 6); on Sunday the timetable is furthest off (13% within two minutes, with buses a median of 13 minutes ahead of it). Bide+ is more accurate than the official real time at 58 of the 60 stops (Figure 5), at every hour of the day (Figure 6, left) and on every day (right). It holds on weekdays (37% against 63% within two minutes, all horizons) and on Sunday (33% against 72%), at the priority stops (35% against 62%) as at the others (37% against 66%), and for the second announced bus, which is further away

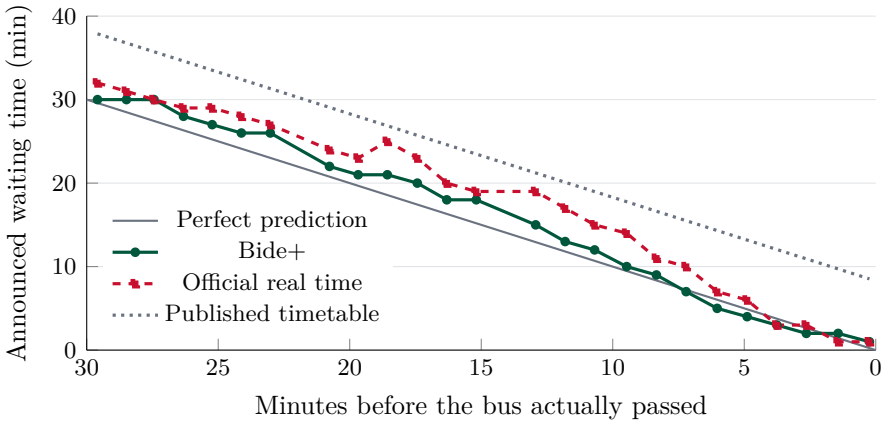


Figure 4: One bus, chosen because both of its error medians are close to the overall ones: line 3641 at stop 270 on Thursday 12 November, queried every minute during the 30 minutes before it passed. The thin line is a perfect prediction. The timetable says the bus is eight minutes later than it really is; the official real time stays several minutes too long until the bus is close; Bide+ follows the line.

(26% against 53%). Both sources are least accurate in the peaks, at 08:00 and around 17:00–19:00, when traffic is least predictable. Table 3 gives the figures by group.

6.5. Robustness

Buses of the same line can bunch, and then the matching could pick the wrong bus. Removing the pairs in which another bus of the same line passed within two minutes of the matched one removes 1.3% of the pairs and changes no figure by more than one point. The day-by-day results (Table 4) show the same pattern on each day.

6.6. The strike day

On Wednesday 11, with minimum services, the official service was within two minutes in 36% of cases and Bide+ in 63%, the same as on a normal day. Bide+ was, however, more often early on that day: in 27% of cases the bus arrived more than two minutes before its prediction, against 12–16% on normal days. Its learned travel times come from normal traffic, and with fewer buses and lighter traffic the buses ran faster than learned.

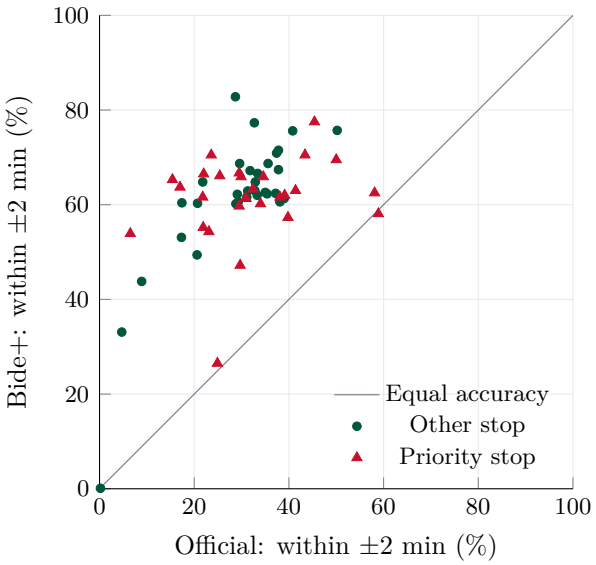


Figure 5: Each point is one of the 60 stops: share of predictions within two minutes, with the bus 3 to 30 minutes away. Bide+ is more accurate at 58 of the 60 stops (above the diagonal).

7. Discussion

7.1. What the comparison shows

On the buses and stops compared, the published timetable is a poor guide to when the bus will come, the operator's real time removes part of its error, and a method that uses only the operator's open data predicts arrivals more accurately than the operator's own service from three minutes before arrival onwards. The gap is largest, about 34 points in the share within two minutes, when the bus is 6 to 20 minutes away: the range in which a passenger decides when to leave for the stop. In that range, a passenger who follows the official waiting time finds that the bus has already gone, or is about to, in roughly half of the cases; with Bide+, in about one in six. Below three minutes both are accurate, because both effectively see the bus approaching.

The comparison does not reveal how the official service computes its waiting times, and we do not claim to know. Its errors grow fast with the horizon and are biased towards announcing the bus later than it comes, the behaviour one would expect from a prediction that does not learn how fast buses actually travel on each segment. Learning those segment times

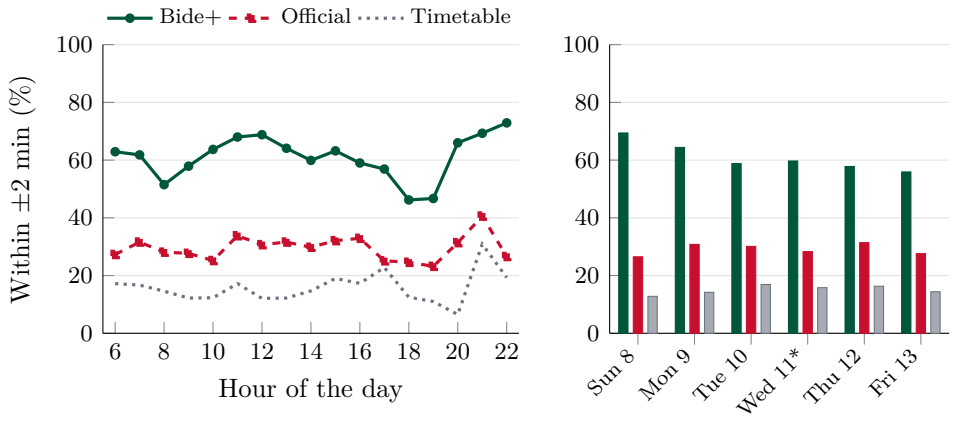


Figure 6: Share within two minutes, bus 3 to 30 minutes away. Left: by hour of the query (five normal days). Right: by day (* general strike, not part of the week's results).

Table 3: Errors by type of day, type of stop and announced bus, all horizons up to 30 minutes. Columns as in Table 2.

		Timetable			Official real time			Bide+		
Group	Pairs	Med.	±2	Early	Med.	±2	Early	Med.	±2	Early
Weekday	135,226	438	16	57	186	37	40	84	63	13
Weekend	24,820	772	13	74	219	33	55	70	72	16
Priority stops	62,750	511	15	60	201	35	41	86	62	14
Other stops	97,296	447	16	59	185	37	43	78	66	14
Next bus (e1)	152,543	472	16	59	188	37	41	80	65	14
Second bus (e2)	7,503	477	14	66	251	26	57	110	53	14

is the core of the Bide+ method (Garrido 2026a; Garrido 2026b), and the previous evaluations against the timetable already showed that the published intermediate times run slow on this network.

7.2. Third-party apps

Many passengers use general-purpose journey planners, which receive real-time data from operators. If such an app shows the operator's prediction unchanged, its accuracy is the official accuracy measured here. We have not measured any third-party app directly, and their processing may differ.

7.3. Limitations

- **Which buses are compared.** An announcement can only be compared if the actual passage of the matched bus can be reconstructed

Table 4: Errors by day, all horizons up to 30 minutes. Columns as in Table 2. Wednesday 11 (general strike) is shown for reference and is not part of the results of the week.

		Timetable			Official real time			Bide+		
Day	Pairs	Med.	±2	Early	Med.	±2	Early	Med.	±2	Early
2026-11-08	24,820	772	13	74	219	33	55	70	72	16
2026-11-09	31,925	448	15	59	179	38	40	75	67	15
2026-11-10	38,704	453	17	56	186	37	39	84	63	12
2026-11-11 (strike)	24,331	562	16	65	200	36	48	84	63	27
2026-11-12	34,293	410	17	56	179	38	38	84	61	12
2026-11-13	30,304	450	15	57	205	34	41	92	59	16

from the feed. This excludes buses for which the stop is at or near the end of their trip (the feed stops reporting them when the trip ends) and buses not reported by the feed. Of the official announcements of the next bus up to 25 minutes ahead, 19% could be paired. The results describe buses whose passage is observable, mostly buses in the middle of their trip, not every announcement.

- **One week, one network.** The data cover five normal days (one of them a Sunday) in November, at 60 stops chosen for their traffic and difficulty. Other seasons, timetables or networks may differ.
- **Resolution of the actual passage.** The feed changes every 60 seconds, so the actual passage is known to about ±30 s. This noise is shared by both sources and cannot explain differences of minutes.
- **Matching.** The official service does not identify the vehicle it announces. Matching by order of passage is what a passenger experiences, and the results do not change when ambiguous cases are removed.
- **The replay, not the app.** Bide+ is recomputed with its reference implementation and tables frozen before the week. The live app recalibrates every night and remembers buses that have just passed, so it may perform slightly differently, most likely better.

8. Reproducibility

The capture program, the list of stops with the reason for each, the days used (`dias.json`), the comparison (`comparar.py`), the tables (`resultados.py`) and the figure data (`figuras.py`) are in the folder `articulo-comparacion` of the repository. The raw captures (feed copies,

official responses and the timetable of each day) are assets of the release `comparacion-2026-11`. The reference implementation of the Bide+ method, its test vectors and the learned tables are those of the companion paper, and the technical report (Garrido 2026a) documents every constant.

9. Conclusions

For one week, at 60 stops of the Bizkaibus network, we recorded the official real-time waiting times minute by minute and compared them, bus by bus, with the arrivals predicted by the Bide+ method, which uses only the operator's published positions and timetable. Over all horizons up to 30 minutes, the published timetable is within two minutes of the actual passage in 16% of cases, the official real time in 36% and Bide+ in 64%. With the bus 6 to 20 minutes away, the official real time is within two minutes in about a third of cases and Bide+ in about two thirds, at almost every stop, hour and day. The official errors are mostly in the direction that makes passengers miss the bus. Operators that publish their vehicle positions already make possible predictions better than their own. Adopting a method that learns segment travel times, or opening their predictions to independent evaluation, would directly benefit their passengers.

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